Form D

Core Focus on Proportions & Probability

Block 4 Review ~ Probability and Random Sampling

ANSWER KEY

Name _____ Period _____ Date _____

Part I – Selected Response

1. At archery practice, Leo hit the target with 18 of his 24 arrows. What is the experimental probability, written as a percent, that his next arrow will hit the target?

A. 18% B. 24% C. 60% D. 75%

D ($\frac{18}{24} = \frac{3}{4} = 0.75 = 75\%$)

2. A spinner has 8 equal sections: 4 red, 3 blue and 1 green. It is spun 40 times. For 2a – 2c, circle YES or NO to show whether each statement is true.

2a. You would expect about 15 spins to land on blue. **YES NO**

2b. You would expect about 10 spins to land on green. **YES NO**

2c. The spinner is certain to land on red exactly 20 times. **YES NO**

2a YES ($\frac{3}{8} \cdot 40 = 15$) · 2b NO ($\frac{1}{8} \cdot 40 = 5$) · 2c NO ($\frac{4}{8} \cdot 40 = 20$ is only the expected number; it is not guaranteed)

3. Lena asked 45 students how they like to spend Saturday. The results are in the table. She wants to show the data in a pie chart. How many degrees should the sector for Sports be?

Activity	Sports	Video games	Reading
Students	18	15	12

A. 144° B. 40° C. 18° D. 162°

A ($\frac{18}{45} = \frac{x}{360}$; $45x = 6{,}480$, $x = 144°$. Check: Video games 120° + Reading 96° + 144° = 360°.)

4. June has four letter cards: W, I, N and D. She mixes them up and draws them one at a time, keeping each card after she draws it. What is the probability she draws them in the order W, I, N, D?

A. $\frac{1}{4}$ B. $\frac{1}{16}$ C. $\frac{1}{24}$ D. $\frac{1}{256}$

C (cards are not replaced: $4 \cdot 3 \cdot 2 \cdot 1 = 24$ possible orders, and only one spells WIND)

5. Ty has cards numbered 1 – 8. He picks a card, puts it back, and then picks a second card. Which of the theoretical probabilities are accurate? Circle all that apply.

A. P(3 then 3) = $\frac{1}{16}$

B. P(even number then odd number) = $\frac{1}{4}$

C. P(9 then 1) = $\frac{1}{72}$

D. P(number less than 3 then 8) = $\frac{1}{32}$

E. P(odd number then 7) = $\frac{1}{16}$

B, D, E. (A: $\frac{1}{8} \cdot \frac{1}{8} = \frac{1}{64}$, not $\frac{1}{16}$; B: ½ · ½ = ¼; C is 0, there is no card 9; D: $\frac{2}{8} \cdot \frac{1}{8} = \frac{2}{64} = \frac{1}{32}$; E: $\frac{4}{8} \cdot \frac{1}{8} = \frac{4}{64} = \frac{1}{16}$)

6. Seven students entered a poster contest. Ribbons are given for first, second and third place. In how many different ways can the three ribbons be awarded?

A. 210 B. 21 C. 343 D. 5,040

A ($7 \cdot 6 \cdot 5 = 210$; a student can't win two ribbons, so it is not $7 \cdot 7 \cdot 7 = 343$)

7. About $\frac{1}{3}$ of the boxes of a cereal have a prize inside. Which simulation is a good model for buying one box?

A. Flip a coin. Heads means a prize.

B. Roll a number cube. A 1 or 2 means a prize; 3 – 6 means no prize.

C. Roll a number cube. A 1 means a prize; 2 – 6 means no prize.

D. Draw from a bag with 3 slips marked "prize" and 1 slip marked "no prize."

B (P(1 or 2) = $\frac{2}{6} = \frac{1}{3}$; A gives ½, C gives $\frac{1}{6}$, D gives ¾)

8. Owen wants to know whether the people in his town want more soccer fields. He asks 50 fans as they leave a soccer game. Why is this sample biased?

A. Fifty people is too many to ask.

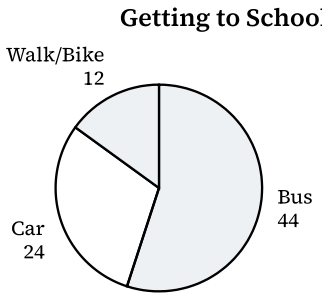
B. Some of the fans might not answer.

C. Soccer fans are more likely to want soccer fields, so the sample does not represent everyone in town.

D. The survey should have been done in the morning.

C (only people at a soccer game could be chosen, so not every resident had an equal chance)

9. Eighty students were chosen at random from a school of 600 and asked how they get to school. The results are below. Which inference is the strongest?



A. Almost all students at the school ride the bus.

B. Most students at the school ride the bus.

C. Most students at the school come by car or walk or bike.

D. Exactly 15% of the students at the school walk or bike.

B (bus is 44 of 80 = 55%, more than half. A overstates 55%; C is only 36 of 80 = 45%; a sample can't tell you an exact percent for the whole school)

10. Tess said the median of the data below is 4, because 4 is the middle number in the list. Which statement is true?

15, 6, 11, 4, 9, 12, 8

A. Tess is correct.

B. The median is about 9.3, because you add the values and divide by 7.

C. There is no median, because no value repeats.

D. Tess did not put the data in order first. The median is 9.

D (in order: 4, 6, 8, 9, 11, 12, 15. About 9.3 is the mean, not the median.)

11. The number of library books Lily and Noah checked out in each of seven weeks is shown below. Which statement is true?

Lily: 6, 9, 4, 9, 7, 3, 9
Noah: 5, 8, 8, 2, 6, 5, 9

A. Lily's data has one mode and Noah's has two modes.

B. Both data sets have a range of 6.

C. Lily's median is 2 more than Noah's.

D. Noah's data has no mode.

A (in order, Lily: 3, 4, 6, 7, 9, 9, 9 → mode 9, range 9 − 3 = 6, median 7. Noah: 2, 5, 5, 6, 8, 8, 9 → modes 5 and 8, range 9 − 2 = 7, median 6.)

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Part II – Constructed Response

Ivy writes each letter of the word BOOKKEEPER on its own card and draws one card at random. Find each probability. Write it as a fraction in simplest form and as a percent.

1. P(K)

$\frac{2}{10} = \frac{1}{5} = 20\%$

2. P(vowel)

$\frac{5}{10} = \frac{1}{2} = 50\%$ (O, O, E, E, E)

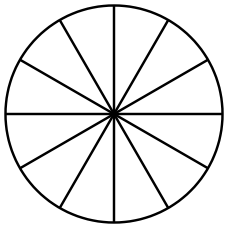
3. P(Z)

0 = 0% (impossible, there is no Z)

4. P(not E)

$\frac{7}{10} = 70\%$ $(1 - \frac{3}{10})$

5. The spinner has 12 equal sections. Color it so that P(red) = $\frac{1}{4}$ and P(blue) = $\frac{1}{3}$. Color the rest yellow. How many sections are each color, and what is P(yellow)?



Red: $\frac{1}{4} \cdot 12 = 3$ sections. Blue: $\frac{1}{3} \cdot 12 = 4$ sections. Yellow: $12 - 3 - 4 = 5$ sections, so P(yellow) = $\frac{5}{12}$.

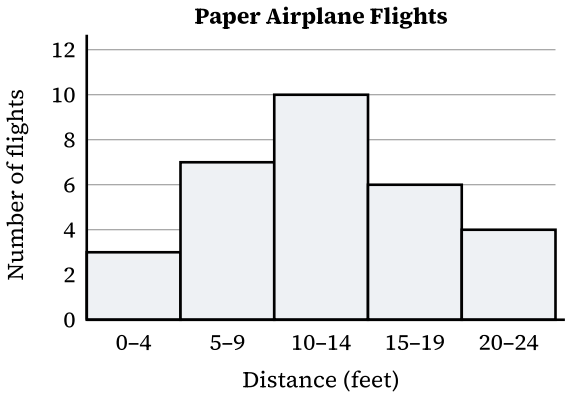
6. A park worker wanted to estimate the number of rabbits in a park. She caught 18 rabbits, tagged them and let them go. A month later she counted 30 rabbits, and 4 of them had tags. Estimate the number of rabbits in the park. Show all work necessary to justify your answer.

135 rabbits. $\frac{4 \text{ tagged}}{30 \text{ counted}} = \frac{18 \text{ tagged}}{n \text{ total}}$; $4n = 540$, $n = 135$. (Check: $\frac{4}{30} = \frac{2}{15}$ and $\frac{18}{135} = \frac{2}{15}$.)

7. Before the class election, Jaden asked a random sample of 60 seventh graders who they will vote for. Twenty-seven chose Kim and 33 chose Lopez. There are 380 seventh graders who will vote. (a) Find P(a student votes for Kim). (b) Predict how many votes Kim will get. (c) Would you predict that Kim wins? Explain.

(a) $\frac{27}{60} = \frac{9}{20}$ (45%). (b) $\frac{9}{20} \cdot 380 = 171$ votes. (c) No. Lopez is predicted to get $380 - 171 = 209$ votes, which is more than half.

8. The histogram shows how far the paper airplanes flew at a science fair contest.



a. How many flights are shown?

3 + 7 + 10 + 6 + 4 = 30 flights

b. Find the probability that a randomly chosen flight went 10 to 19 feet.

$\frac{10+6}{30} = \frac{16}{30} = \frac{8}{15}$

c. If 150 more airplanes are flown, predict how many will fly 10 to 14 feet.

$\frac{10}{30} = \frac{1}{3}$; $\frac{1}{3} \cdot 150 = 50$ airplanes

9. Nico picks an outfit with one shirt (polo or tee), one pair of pants (jeans, khakis or shorts) and one pair of shoes (sneakers or sandals). How many different outfits are possible? If he picks an outfit at random, what is the probability it has shorts and sandals? Show a tree diagram, list or table to support your answer.

12 outfits ($2 \cdot 3 \cdot 2$). Polo: jeans–sneakers, jeans–sandals, khakis–sneakers, khakis–sandals, shorts–sneakers, shorts–sandals; tee: the same 6. Shorts and sandals: polo–shorts–sandals and tee–shorts–sandals, so $P = \frac{2}{12} = \frac{1}{6}$.

10. A bag has 5 red, 3 yellow and 2 green beads. (a) Hana picks a bead, puts it back, and picks again. What is P(yellow then yellow)? (b) Hana picks a bead, keeps it, and picks again. What is P(yellow then yellow)? (c) Which way gives the better chance of two yellow beads?

(a) $\frac{3}{10} \cdot \frac{3}{10} = \frac{9}{100}$. (b) $\frac{3}{10} \cdot \frac{2}{9} = \frac{6}{90} = \frac{1}{15}$ (only 2 yellow and 9 beads are left). (c) Putting it back: $\frac{9}{100} = 0.09$ is greater than $\frac{1}{15} \approx 0.067$.

11. Max guesses on every question of a 4-question true/false quiz. (a) Describe how to use a coin to simulate one quiz. (b) What is the theoretical probability he gets all 4 right? (c) His class ran the simulation 80 times, and 6 trials had all 4 right. What is the experimental probability?

(a) Sample: Let heads = a correct guess and tails = a wrong guess. Toss the coin 4 times for one quiz, and record whether all 4 were heads. Repeat for many trials. (b) $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{16}$. (c) $\frac{6}{80} = \frac{3}{40}$. It is close to $\frac{1}{16}$, and more trials would usually bring it closer.

12. A bakery made 600 muffins one morning. The baker checked every 25th muffin as it came out of the oven, so she checked 24 muffins. One of them was underbaked. She predicts that about 25 of the 600 muffins are underbaked. Is her prediction as accurate as possible based on the sample? Explain your reasoning.

Yes. Checking every 25th muffin spreads the sample over the whole morning's baking, so it represents all 600 muffins. $\frac{1}{24} \cdot 600 = 25$ underbaked muffins.

13. Two different random samples were used to find students' favorite school subject. The results are below. Make at least two inferences. Then decide whether "Most students prefer PE" is a strong or weak inference.

	Math	Science	English	PE	TOTAL
Sample 1	18	12	6	24	60
Sample 2	22	16	10	32	80

Sample answers: PE is the favorite subject (40% in both samples). English is the least favorite (10% and 12.5%). Math is second (30% and 27.5%). "Most students prefer PE" is weak: only 40% chose PE, which is less than half.

Use the data below to answer numbers 14 – 20. The data shows the number of minutes it took 9 students in each group to finish a puzzle.

Group X: 14, 19, 12, 44, 14, 17, 13, 15, 14
Group Y: 23, 14, 26, 19, 28, 17, 21, 25, 16

14. Find the mean of each data set.

Group X: $\frac{162}{9}$ = 18 minutes. Group Y: $\frac{189}{9}$ = 21 minutes.

15. Find the median and mode of each data set.

Group X: 12, 13, 14, 14, 14, 15, 17, 19, 44 → median 14, mode 14. Group Y: 14, 16, 17, 19, 21, 23, 25, 26, 28 → median 21, no mode.

16. Gabe said the mean, 18 minutes, is the best number to describe a typical time in Group X. Is he right? Explain your reasoning.

No. The 44 is an outlier that pulls the mean up: 7 of the 9 times are below 18. The median, 14 minutes, describes a typical time better.

17. Find the five-number summary of each data set.

Group X: 12 ~ 13.5 ~ 14 ~ 18 ~ 44. Group Y: 14 ~ 16.5 ~ 21 ~ 25.5 ~ 28. (Leave out the median: Q1 is the median of the lowest 4 values, Q3 of the highest 4.)

18. Find the range and the interquartile range of each data set.

Group X: range 44 − 12 = 32; IQR 18 − 13.5 = 4.5. Group Y: range 28 − 14 = 14; IQR 25.5 − 16.5 = 9.

19. Find the mean absolute deviation of each data set.

Group X (mean 18): distances 4, 1, 6, 26, 4, 1, 5, 3, 4; sum 54; MAD = $\frac{54}{9}$ = 6 minutes. Group Y (mean 21): distances 2, 7, 5, 2, 7, 4, 0, 4, 5; sum 36; MAD = $\frac{36}{9}$ = 4 minutes.

20. Write one sentence comparing a measure of variability between the two data sets.

Sample: Group X has the larger range (32 vs. 14) and MAD (6 vs. 4) because of the outlier 44, but Group Y has the larger IQR (9 vs. 4.5), so the middle half of Group Y's times is more spread out.

Core Focus on Proportions & Probability

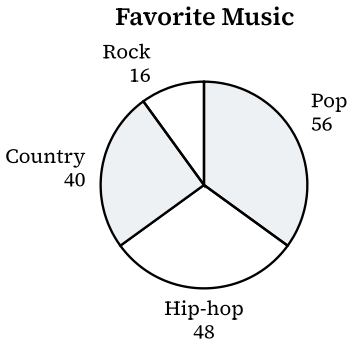
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Part III – Problem Solving

Diego asked a random sample of 160 students at his school what kind of music they like best and how they usually listen to it. The results are below.



Listens On	Number of Students
Phone	96
Speaker	40
Radio	24

There are 1,000 students at Diego's school. How many degrees is the Hip-hop sector of the pie chart? Based on Diego's data, what is the probability that the next student he asks will like hip-hop best and listen on a speaker? How many students in the whole school would you predict like hip-hop best and listen on a speaker?

Sector: $\frac{48}{160} = \frac{x}{360}$, so $x = 108°$. P(hip-hop) = $\frac{48}{160} = \frac{3}{10}$. P(speaker) = $\frac{40}{160} = \frac{1}{4}$. P(hip-hop and speaker) = $\frac{3}{10} \cdot \frac{1}{4} = \frac{3}{40}$. Prediction: $\frac{3}{40} \cdot 1{,}000 = 75$ students.